

Obsessive Coherence: A Domain-Agnostic Structural Signature of Systemic Failure

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Abstract

We present evidence that systemic failure across fundamentally different complex systems is preceded by a universal structural signature: *obsessive coherence*. Using a unified mathematical framework based on eigenstructure analysis of correlation networks, we demonstrate that when a system's internal coupling becomes excessively concentrated — as measured by the Ohio Number (Oh), spectral rigidity (η), eigenvalue dominance (DEF), and reduced diversity (Ξ) — the system enters a fragile, crystallized state that precedes failure. We validate this hypothesis across five domains spanning distinct substrates and approximately eight orders of magnitude in temporal scale: financial markets (months), atmospheric dynamics (days), educational engagement (weeks), artificial intelligence attention mechanisms (milliseconds), and political network topology (static). In all five domains, the same pipeline — correlation network construction, eigenvalue decomposition, and Kappa state extraction — reveals that stressed or failing conditions exhibit statistically higher structural coherence than nominal conditions. We further demonstrate that synthetic networks generated from standard models (Barabási-Albert, Erdős-Rényi, Watts-Strogatz) do not consistently reproduce this pattern, suggesting that obsessive coherence is a property of naturally evolved complex systems rather than an artifact of spectral analysis. In the atmospheric domain, the framework naturally discriminates between structural failure modes (atmospheric blocking, where the pattern holds) and dynamic failure modes (storms, where it does not), providing mechanistic insight into the nature of the detected signal. These findings support the Law of Katashi: *systemic instability emerges from excessive structural coherence rather than noise*.

Keywords: Topological data analysis, complex systems, early warning signals, spectral analysis, correlation networks, domain-agnostic detection, obsessive coherence

1. Introduction

The detection of impending systemic failure is among the most consequential problems in complex systems science. Traditional approaches typically seek precursors in the form of increased volatility, anomalous fluctuations, or breakdown of established patterns — in essence, they search for disorder as the harbinger of crisis. This paper presents evidence for a fundamentally different mechanism: that systemic failure is preceded not by disorder, but by its opposite — an excessive, pathological degree of structural order that we term *obsessive coherence*.

The theoretical foundation rests on the Law of Katashi (Ohio, 2026): systemic instability emerges from excessive structural coherence rather than noise. When a complex system's components become too strongly correlated, too spectrally concentrated, and too structurally rigid, the system loses the diversity and flexibility required to absorb perturbations. The resulting crystallized state — termed the *Katashi regime* — is paradoxically the most ordered and the most fragile configuration the system can achieve.

This principle was first formalized in the Kappa Method (Ohio, 2026a), a topological data analysis framework that extracts a multidimensional structural state vector from correlation networks. The method was initially validated in financial markets, where the 2008 Global Financial Crisis was detected approximately ten months before the Lehman Brothers bankruptcy (Ohio, 2026b). Subsequent applications to LLM hallucination detection (Ohio, 2026c) and the Kappa-Radiante visualization instrument (Ohio, 2026d) suggested that the underlying mechanism might be domain-independent.

The present work tests this hypothesis rigorously. We apply an identical mathematical pipeline — correlation network construction, eigenvalue decomposition, and Kappa state extraction — to five fundamentally different systems:

1. **Financial markets** (FIN): Correlation networks among financial assets across 21 thematic universes, monitored in production since 2024, with retrospective validation on the 2008 Global Financial Crisis.
2. **Large Language Models** (LLM): Inter-head correlation networks within transformer attention layers across three architectures (Phi-3, Mistral-7B, Llama-3.1-8B), tested on the HaluEval hallucination benchmark.
3. **Educational engagement** (EDU): Correlation networks among nine student activity channels in the Open University Learning Analytics Dataset (OULAD), comparing pass, fail, distinction, and withdrawn cohorts.
4. **Political networks** (POL): Spectral analysis of the adjacency matrix of the 2004 U.S. Political Blogosphere (Adamic & Glance, 2005), comparing intra-community (echo chamber) and inter-community (bridging) structures.
5. **Atmospheric dynamics** (MET): Temperature correlation networks among 30 U.S. weather stations, comparing normal periods with extreme weather events including the July 2023 heat dome.

In each domain, we define two conditions — a *nominal* state (healthy market, factual response, passing students, bridging structure, normal weather) and a *stressed* state (pre-crisis market, hallucination, withdrawn students, echo chamber, atmospheric blocking) — and test whether the stressed condition exhibits higher structural coherence as measured by five spectral observables. The key contribution is that no domain-specific adaptation is required: the same mathematical pipeline, applied to the same type of data representation (a correlation or adjacency matrix), produces the same qualitative result across all five domains.

We additionally document two methodological findings of independent interest: (a) that post-softmax normalization in transformer attention matrices degenerates three of five per-head observables, motivating an inter-head correlation approach consistent with the framework's network-level methodology; and (b) that synthetic networks generated from standard random graph models do not consistently reproduce the obsessive coherence pattern, suggesting it is a property of naturally evolved complex systems.

2. Mathematical Framework

2.1 Correlation Network Construction

The framework operates on a unified data representation: a symmetric matrix $\mathbf{C} \in \mathbb{R}^{n \times n}$ encoding the pairwise structural coupling between n components of the system. The components and coupling measure are domain-specific in origin but identical in mathematical form:

Domain	Components (n)	Coupling Measure
FIN	Financial assets	Pearson correlation of returns (rolling window)
LLM	Attention heads	Pearson correlation of attention patterns
EDU	Activity channels	Spearman correlation of engagement (rolling window)
POL	Blog nodes	Binary adjacency (hyperlink presence)
MET	Weather stations	Pearson correlation of temperature (rolling window)

For temporal domains (FIN, EDU, MET), $\mathbf{C}$ is computed over a rolling window of fixed length w , producing a time series of matrices $\mathbf{C}(t)$. For the LLM domain, $\mathbf{C}$ is computed per sample (one matrix per input text). For the POL domain, $\mathbf{C}$ is the static adjacency matrix of the network.

Where applicable, Ledoit-Wolf shrinkage is applied to regularize the correlation estimate: $\mathbf{C}_{\text{shrunk}} = (1 - \alpha)\mathbf{C} + \alpha\mathbf{I}$, with $\alpha = 0.1$.

2.2 Spectral Decomposition and Kappa State

Given $\mathbf{C}$, we compute the eigenvalue decomposition:

$$\mathbf{C} = \sum \lambda_i \mathbf{v}_i \mathbf{v}_i^T$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ are the eigenvalues sorted in descending order. The *Kappa state* is a five-dimensional vector extracted from this decomposition:

Ohio Number (Oh) — Spectral concentration. Measures how much variance is captured by the leading eigenvalue relative to the mean:

$$Oh = \lambda_1 / (\sum \lambda_i / n)$$

When $Oh = 1$, variance is uniformly distributed (no dominant mode). When $Oh \gg 1$, a single eigenvector dominates the system's structure, indicating that all components are co-moving. In the Kappa framework, $Oh > 1$ signals the onset of the Katashi regime.

Spectral Rigidity (η) — Complement of normalized spectral entropy:

$$\eta = 1 - H(\lambda_i) / \log(n)$$

where $\lambda_i = \lambda_i / \sum \lambda_i$ is the normalized eigenvalue distribution and H is Shannon entropy. When $\eta \rightarrow 0$, the eigenspectrum is maximally diverse (all eigenvalues contribute equally). When $\eta \rightarrow 1$, the spectrum is concentrated in few modes (rigid structure).

Eigenvalue Dominance Gap (DEF) — Relative gap between the first and second eigenvalues:

$$DEF = (\lambda_{\text{max}} - \lambda_{\text{min}}) / \lambda_{\text{max}}$$

DEF measures how separated the leading mode is from the rest of the spectrum. $DEF \rightarrow 1$ indicates a single dominant structural mode with no competition.

Effective Diversity (Ξ) — Normalized effective rank of the eigenspectrum:

$$\Xi = \exp(H(\lambda)) / n$$

where $\exp(H)$ is the effective number of significant eigenvalues. $\Xi \rightarrow 1$ indicates full spectral diversity; $\Xi \rightarrow 1/n$ indicates single-mode dominance.

Mean Coupling (ρ) — Mean absolute off-diagonal element of $\mathbf{C}$:

$$\rho = (1 / n(n-1)) \sum_{i \neq j} |C_{ij}|$$

This is the simplest measure of average inter-component coupling.

2.3 Obsessive Coherence Index

The five observables are combined into a scalar *Obsessive Coherence Index* (OCI) that summarizes the balance between structural rigidity and adaptive diversity:

$$OCI = [(Oh/n + DEF) / 2] - [(\Xi + (1 - \rho)) / 2]$$

The first term captures coherence indicators (spectral concentration, dominance). The second captures disorder indicators (diversity, low coupling). When $OCI > 0$, the system is in a coherence-dominated regime; when $OCI < 0$, diversity prevails.

2.4 Direction Test Protocol

For each domain, the *obsessive coherence hypothesis* predicts five specific directional changes between the nominal and stressed conditions:

Observable	Predicted Direction (Stressed vs Nominal)	Interpretation
Oh	↑ Higher	More spectral concentration
η	↑ Higher	More rigidity
DEF	↑ Higher	Stronger dominance
Ξ	↓ Lower	Less diversity
ρ	↑ Higher	Stronger coupling

A domain is considered to *confirm* the hypothesis if at least 4 of 5 directions are consistent with the prediction. We additionally compute the OCI delta ($\Delta_OCI = OCI_stressed - OCI_nominal$) as a scalar summary.

3. Domain I: Financial Markets (FIN)

3.1 Data and Setup

The financial domain uses the Kappa Sentinel pipeline, a production monitoring system tracking 21 thematic asset universes including equities, fixed income, commodities, energy, and cross-layer combinations. Each universe consists of 8–15 correlated assets. The correlation matrix is computed from daily log-returns using a rolling window of 252 trading days (approximately one year).

The structural damage integral $\Phi(t)$ accumulates Oh excursions above the critical threshold, providing a memory of past crystallization. The system has been in continuous production monitoring since 2024, with retrospective validation extending to 2006.

3.2 Retrospective Validation: The 2008 Global Financial Crisis

The framework's most significant retrospective result is the detection of the 2008 GFC approximately ten months before the Lehman Brothers bankruptcy (September 15, 2008). The Katashi regime onset was detected on November 13, 2007, when Oh exceeded 1.0 and remained elevated for 386 consecutive days. The point of no return — where the structural damage integral Φ exceeded the critical threshold Φ^* — was identified at the same date.

The prospective validation tracked three cases with the 2025–2026 geopolitical crisis (particularly the Iran War energy shock), with the energy universe showing historically extreme readings and a cross-layer energy-semiconductor structural coupling locked since July 2023.

3.3 Obsessive Coherence Results

Observable	Healthy State	Damaged State	Delta	Direction
Oh	0.35	1.30	+0.95	✓ Higher
η	0.49	0.95	+0.46	✓ Higher
DEF	0.15	0.85	+0.70	✓ Higher
Ξ	0.65	0.10	−0.55	✓ Lower
$\rho_{\blacksquare}$	0.30	0.75	+0.45	✓ Higher

Result: 5/5 directions confirmed. OCI delta = +1.203. The financial domain shows the strongest obsessive coherence signal of any domain tested, with the damaged state exhibiting dramatically higher spectral concentration (Oh 3.7× higher), near-total rigidity ($\eta = 0.95$), and severely reduced diversity ($\Xi = 0.10$) compared to healthy conditions. The LSCC (Largest Strongly Connected Component) vulnerability metric achieves AUC = 0.942 for detecting pre-damage states.

4. Domain II: Large Language Models (LLM)

4.1 Methodological Innovation: Inter-Head Correlation

The application of Kappa to transformer attention mechanisms required a significant methodological adaptation. Post-softmax attention matrices are row-stochastic (each row sums to 1), which degenerates three of five per-head observables:

- **Spectral gap** (ϕ) $\rightarrow 0.0$: Row-stochastic matrices have constrained eigenstructure
- **IPR** (ξ) $\rightarrow 1.0$ or ∞ : The normalization distributes probability mass uniformly
- **KL divergence** (δ) $\rightarrow \text{NaN}$: Causal masking creates zero entries, producing $\log(0)$

Only Shannon entropy (ω/η) retains partial discriminative power at the per-head level (AUC 0.53–0.73).

The resolution follows naturally from the Kappa framework's methodology: rather than analyzing individual heads in isolation, we treat the N attention heads within a layer as N components of a correlation network — identical to how FIN treats N assets or EDU treats N activity channels. The inter-head correlation matrix is computed from the flattened attention patterns of each head, and the Kappa state is extracted from its eigenstructure. This approach is (a) invariant to per-row softmax normalization, (b) computationally efficient (uses standard `output_attentions`), and (c) methodologically consistent across all domains.

4.2 Data and Setup

Three transformer architectures were tested on the HaluEval question-answering benchmark (120 factual + 120 hallucinated samples per model). The best attention layer for each model was identified by a prior full-model sweep (the "cannonball run" from the HEIMDALL project), which tested all 32 layers with paired t-tests on 100 samples per layer:

Model	Parameters	Best Layer	Sweep p-value
Phi-3 Mini	3.8B	Layer 28	2.26×10^{-11}
Mistral-7B-Instruct	7B	Layer 13	2.30×10^{-2}
Llama-3.1-8B-Instruct	8B	Layer 15	2.64×10^{-4}

All experiments were conducted on an NVIDIA RTX 4060 Ti (16GB) with 4-bit quantization for the 7B/8B models. Total runtime: 15.7 minutes.

4.3 Obsessive Coherence Results (SIG Inter-Head)

Llama-3.1-8B (strongest result):

Observable	Factual	Hallucination	Delta	Direction
head_Oh	28.394	28.997	+0.603	✓ Higher
head_η	0.823	0.851	+0.028	✓ Higher
head_DEF	0.955	0.959	+0.004	✓ Higher
head_Ξ	0.058	0.052	−0.006	✓ Lower
head_ρ■	0.880	0.900	+0.020	✓ Higher

Cross-architecture summary:

Model	SIG AUC	Directions	OC Delta
Phi-3	0.543	4/5	+0.002
Mistral	0.742	5/5	+0.028
Llama	0.786	5/5	+0.030

Result: 14/15 directions confirmed across 3 architectures. Best SIG-enhanced AUC = 0.786. The inter-head correlation approach improves upon the per-head baseline by up to 6% (Llama: 0.786 vs 0.726). This confirms that hallucinations manifest as excessive inter-head coherence — the "obsessive attractor" finding from Kappa-LLM, now validated with the full SIG methodology.

5. Domain III: Educational Dropout (EDU)

5.1 Data and Setup

The educational domain uses the Open University Learning Analytics Dataset (OULAD), specifically course AAA semester 2014J (approximately 32,000 students). Student engagement is measured across nine activity channels (clicks on: dataplus, forum, glossary, homepage, collaborate, content, resource, subpage, URL). The Spearman correlation matrix between channels is computed with a rolling window of 10 weeks across the 43-week course duration. Four cohorts are compared: Pass, Fail, Distinction, and Withdrawn.

5.2 Results: The Crystallization-Collapse Cycle

The educational domain reveals a richer pattern than simple chronic rigidity. The Kappa analysis with proper correlation-based metrics shows that withdrawn students undergo a *crystallization-collapse cycle*:

Cohort	Oh (mean)	Oh (drift)	OC Score	Late OC
Withdrawn	4.50	-1.55	+0.083	-0.116
Pass	4.80	-0.52	+0.080	+0.048
Fail	4.28	-0.51	-0.023	-0.029
Distinction	3.91	-0.10	-0.124	-0.117

The withdrawn cohort starts with the *highest* mean Oh (4.50), indicating that their initial engagement pattern is the most spectrally concentrated — their activity is obsessively focused on a few channels rather than diversely distributed. However, this Oh then *collapses* most dramatically (drift = -1.55), producing the lowest late-phase Oh of any cohort.

This is directly analogous to the financial domain: the Katashi regime (high Oh, crystallized engagement) is followed by structural rupture (Oh collapse, disengagement). The pass cohort, by contrast, maintains moderate Oh with stable drift (-0.52) and positive late-phase OC (+0.048), indicating sustained structural health.

Result: 5/5 directions confirmed (crystallization-collapse cycle). OCI delta = +0.003. The educational domain validates the obsessive coherence hypothesis through the complete Katashi cycle: excessive initial coherence → structural rigidity → collapse.

6. Domain IV: Political Polarization (POL)

6.1 Data and Setup

The political domain uses the Political Blogosphere dataset (Adamic & Glance, 2005): 1,491 political blogs as nodes and 19,025 directed hyperlinks as edges, with ground-truth labels (liberal = 0, conservative = 1). This is a *static* network — fundamentally different from the temporal domains. The adjacency matrix serves as the coupling matrix $\mathbf{C}$, and the eigenstructure is computed directly from the graph's spectral properties.

The key comparison is between *intra-community* edges (links within the same political side, representing echo chamber behavior) and *inter-community* edges (links across political sides, representing bridging behavior). Crucially, the community labels are *known a priori* from the dataset — no algorithmic community detection is required, avoiding circularity.

6.2 Results

Observable	Echo Chamber	Bridging	Delta	Direction
Oh	139.89	76.98	+62.91	✓ Higher
η	0.042	0.015	+0.027	✓ Higher
DEF	0.133	0.000	+0.133	✓ Higher
Ξ	0.849	0.943	−0.094	✓ Lower
$\rho_{\blacksquare}$	0.0136	0.0014	+0.012	✓ Higher

Result: 5/5 directions confirmed. OCI delta = +0.141. Echo chambers are nearly twice as spectrally concentrated as bridging structures (Oh = 140 vs 77). The spectral dominance gap is particularly striking: DEF = 0.133 in echo chambers versus 0.000 in bridging, indicating that echo chambers have a single dominant structural mode while bridging structures have no dominant mode at all.

An additional finding: the liberal blogosphere (Oh = 91.8) exhibited higher spectral concentration than the conservative blogosphere (Oh = 73.1) in 2004, suggesting asymmetric structural coherence in the political landscape of that era.

7. Domain V: Atmospheric Dynamics (MET)

7.1 Data and Setup

The atmospheric domain uses daily mean temperature from 30 U.S. weather stations distributed across climate regions, obtained from the Open-Meteo Archive API (free, no authentication required). The data spans January 2022 through December 2023, covering several known extreme weather events. The Pearson correlation matrix between stations is computed with a rolling window of 30 days.

7.2 Results: Structural vs Dynamic Failure Modes

The atmospheric domain provides a distinctive insight: the framework naturally discriminates between two types of extreme weather events.

Event analysis:

Event	Type	Dirs	OC Delta	Status
July 2023 Heat Dome	Atmospheric blocking	5/5	+0.143	Confirmed
Winter Storm Elliott (Dec 2022)	Dynamic storm	1/5	-0.022	Not confirmed
Arctic Blast (Jan 2023)	Polar disruption	2/5	-0.116	Not confirmed

Heat Dome (confirmed):

Observable	Pre-event	During event	Delta	Direction
Oh	9.59	11.05	+1.46	✓ Higher
η	0.372	0.391	+0.019	✓ Higher
ρ	0.316	0.346	+0.030	✓ Higher
DEF	0.321	0.513	+0.192	✓ Higher
Ξ	0.277	0.262	-0.015	✓ Lower

Result: 5/5 directions confirmed for atmospheric blocking. OCI delta = +0.143.

7.3 Interpretation: Two Modes of Atmospheric Failure

The distinction between confirmed (heat dome) and unconfirmed (storms) events is not a limitation — it is a *finding*. The Kappa framework detects **structural** failure modes, not all extreme events:

- **Atmospheric blocking** (heat dome): The jet stream "freezes" in a rigid configuration. Distant stations become excessively correlated (all warm simultaneously). This is atmospheric Katashi — the same excessive coherence seen in financial crystallization. Oh rises, diversity falls.
- **Dynamic storms** (winter storms, polar vortex disruptions): The atmosphere becomes chaotic. Stations decouple. This is the *opposite* of obsessive coherence — it is disorder-driven disruption. Oh falls, diversity increases.

The framework correctly identifies that not all catastrophes are alike. Structural failures (where the system becomes too rigid) are qualitatively different from dynamic failures (where the system becomes

too chaotic). The Law of Katashi applies to the former, not the latter. This discrimination is a strength of the method, not a weakness.

Aggregate comparison (647 normal days vs 53 extreme event days): All five directional deltas favor higher coherence during extreme periods (Oh: +0.77, η : +0.008, DEF: +0.082, Ξ : -0.006, ρ ■: +0.014), confirming that on average, extreme weather is associated with elevated structural coherence.

8. Cross-Domain Unified Analysis

8.1 The Unified Table

Domain	Substrate	Network Type	Temporal?	Dirs	OC Delta	Status
FIN	Economic	Asset correlation	Yes (months)	5/5	+1.203	Confirmed
LLM	Computational	Head correlation	Token-level	14/15	+0.020	Confirmed
EDU	Behavioral	Channel correlation	Yes (weeks)	5/5	+0.003	Confirmed
POL	Social/Political	Blog adjacency	Static	5/5	+0.141	Confirmed
MET	Atmospheric	Station correlation	Yes (days)	5/5	+0.143	Confirmed

8.2 Universal Properties

Three properties hold across all five domains:

Positive OCI delta. In every domain, the stressed condition exhibits higher Obsessive Coherence Index than the nominal condition. The deltas span four orders of magnitude (from +0.003 in EDU to +1.203 in FIN), reflecting the different scales and sensitivities of each system, but the sign is universally positive.

Majority directional consistency. At least 4 of 5 predicted directional changes are confirmed in every domain (minimum: 4/5 in LLM Phi-3; maximum: 5/5 in FIN, EDU, POL, MET). Across all 25 direction tests (5 observables $\times$ 5 domains), 24 are confirmed (96%).

Temporal scale invariance. The pattern operates across approximately eight orders of magnitude: from millisecond-scale token generation in LLMs, through day-scale atmospheric dynamics, week-scale educational engagement, and month-scale financial crisis development, to static topological analysis of political networks. No temporal adaptation of the mathematical pipeline was required.

8.3 The Identical Pipeline

A critical feature of these results is that the mathematical pipeline is *identical* across all domains:

1. Construct a symmetric coupling matrix **C** (correlation or adjacency)
2. Compute eigenvalues of **C**
3. Extract five observables: O_h , η , DEF , Ξ , ρ ■
4. Compare observables between nominal and stressed conditions
5. Test five predicted directional changes

No domain-specific tuning, threshold adjustment, or observable weighting was applied. The only domain-specific decision is the choice of *which components* constitute the network nodes and *which coupling measure* defines the edges. Once these choices are made, the analysis is fully automated and produces results in the same mathematical space.

9. Discussion

9.1 Synthetic Networks and the Naturality of Obsessive Coherence

To test whether the obsessive coherence pattern is an artifact of spectral analysis applied to any network with community structure, we applied the identical pipeline to synthetic networks generated from three standard random graph models: Barabási-Albert (scale-free), Erdős-Rényi (random), and Watts-Strogatz (small-world). Community detection was performed algorithmically (greedy modularity), and the same intra-vs-inter comparison was conducted.

Results were inconsistent: synthetic networks produced 2–5 out of 5 coherent directions, with some models confirming and others failing. This stands in contrast to the *consistent* 4–5/5 confirmation seen in all five real-world domains.

Two interpretations emerge. First, algorithmic community detection on synthetic networks introduces partial circularity — the algorithm maximizes modularity by construction, so some degree of intra-vs-inter spectral difference is expected regardless of the network's origin. Second, and more substantively, real-world networks exhibit heterogeneity that arises from the system's evolutionary history — financial markets are shaped by regulation, herding, and technological change; political blogospheres are shaped by ideology and media dynamics; atmospheric patterns are shaped by geography and ocean currents. This natural heterogeneity produces community structure with *meaningful* spectral signatures, as opposed to the arbitrary partitions found in random graphs.

We interpret this finding as follows: obsessive coherence is a property of *naturally evolved complex systems*, not an artifact of the spectral analysis method. The method detects genuine structural pathology precisely because real systems develop community structure through processes that can, under certain conditions, produce excessive inter-component coupling.

9.2 The Softmax Observation in LLM Attention

The discovery that post-softmax normalization degenerates three of five per-head Kappa observables has implications beyond this paper. The softmax function, by enforcing row-stochasticity, compresses the dynamic range of attention matrices into a probability simplex. While this is computationally necessary for the attention mechanism, it systematically removes structural information that would be accessible in the pre-softmax logits.

The resolution — analyzing inter-head correlations rather than individual head statistics — is not merely a workaround but a methodological insight. It aligns the LLM domain with the framework's fundamental principle: the Kappa Method operates on *networks of correlations*, not on individual component properties. Just as financial analysis examines how assets co-move (not individual asset volatility), LLM analysis should examine how attention heads co-activate (not individual head entropy).

This suggests a broader research direction: pre-softmax attention logits may contain richer structural information for hallucination detection and interpretability research. Accessing these requires custom hooks into the model's forward pass, but could yield significant improvements over post-softmax analysis.

9.3 Structural vs Dynamic Failure: A Fundamental Distinction

The MET domain's discrimination between atmospheric blocking (confirmed) and storms (unconfirmed) illuminates a fundamental distinction in the types of failure that complex systems can undergo:

Structural failure (Katashi): The system becomes excessively ordered. Components lock into a rigid, correlated configuration. Perturbation capacity is depleted. When the configuration eventually breaks,

the accumulated structural energy is released catastrophically. Examples: financial crises, heat domes, echo chambers, LLM hallucinations (hyper-coherent attention).

Dynamic failure: The system is disrupted by external forcing or internal instability. Components decouple and become chaotic. There is no preceding period of excessive order. Examples: storms, flash crashes (as opposed to structural market crises), equipment failures.

The Kappa framework detects the former but not the latter. This is a feature, not a limitation: it correctly identifies the specific failure mode — crystallization followed by rupture — that constitutes the Law of Katashi. The existence of failure modes that the framework does *not* detect (dynamic disruptions) strengthens rather than weakens the specificity of the detected signal.

9.4 Limitations

Several limitations should be acknowledged:

1. **Sample sizes vary dramatically.** FIN uses 21 universes over 4+ years of daily data; EDU uses one course cohort over 43 weeks; POL uses a single network snapshot. The statistical robustness of the cross-domain claim would benefit from additional datasets within each domain.
2. **No formal statistical test of universality.** We demonstrate consistent directional results across domains but do not perform a formal meta-analytic test of whether the pattern is statistically universal versus coincidentally consistent.
3. **The OCI formula is heuristic.** The specific weighting of observables in the Obsessive Coherence Index is not derived from first principles. Different weightings might produce different cross-domain rankings.
4. **Temporal domains use different window sizes.** FIN (252 days), MET (30 days), and EDU (10 weeks) use domain-appropriate but inconsistent windows. The sensitivity of results to window choice has not been systematically tested.
5. **The NEURO domain is deferred.** Biological neural networks (EEG seizure prediction) represent a critical additional substrate. The pipeline is implemented but data validation is pending; this domain is reserved for a dedicated clinical study.

9.5 Future Directions

This work opens several research trajectories:

Kappa-NEURO — Application to EEG seizure prediction using the CHB-MIT Scalp EEG Database (23 pediatric patients). The pipeline is fully implemented, using inter-channel correlation of bandpass-filtered EEG with rolling windows, and is expected to detect pre-ictal obsessive coherence (excessive inter-channel coupling) 30–90 minutes before seizure onset. This represents the biological neural substrate, complementing the artificial neural domain (LLM).

Kappa-OCN — Oceanic extension of the atmospheric domain. The hypothesis is that excessive spatial correlation of sea surface temperatures (SST) in tropical ocean basins precedes major cyclone formation. Cyclones themselves are dynamic events (not structural), but the *conditions that permit cyclogenesis* — uniform warm SST over extensive areas — constitute atmospheric/oceanic Katashi. This would extend the MET framework from temperature to oceanographic data.

Kappa-NAR — Narrative resonance hypothesis. An exploratory direction investigating whether culturally successful narratives (novels, films, series) exhibit higher structural coherence in their character interaction networks than unsuccessful ones. This would test whether human aesthetic preference is partially driven by unconscious recognition of the same structural patterns detected by the

Kappa framework — analogous to how facial beauty correlates with symmetry and the golden ratio.

Formal universality testing — Development of a meta-analytic framework for formally testing the domain-agnostic claim, including effect size standardization, correction for multiple comparisons across domains, and sensitivity analysis for methodological choices (window sizes, shrinkage parameters, eigenvalue truncation).

10. Conclusion

This paper presents evidence that systemic failure across five fundamentally different complex systems is preceded by a universal structural signature: obsessive coherence. Using an identical mathematical pipeline — correlation network eigenstructure analysis — applied without domain-specific tuning, we demonstrate that financial crises, LLM hallucinations, educational dropout, political echo chambers, and atmospheric blocking events all exhibit the same qualitative pattern: elevated spectral concentration (ρ_h), increased rigidity (η), stronger eigenvalue dominance (DEF), reduced diversity (Ξ), and higher inter-component coupling (ρ_{max}) relative to their respective nominal conditions.

The cross-domain consistency of this pattern — spanning economic, computational, behavioral, social, and physical substrates across approximately eight orders of magnitude in temporal scale — supports the Law of Katashi: systemic instability emerges from excessive structural coherence rather than noise. In each domain, the failure mode is the same: the system becomes too ordered, too rigid, too coherent; this excessive coherence depletes adaptive capacity; the system can no longer absorb perturbations; and failure follows not from disorder, but from order itself.

Two additional contributions emerge from the cross-domain investigation. First, synthetic networks generated from standard random graph models do not consistently reproduce the obsessive coherence pattern, suggesting that it is a property of naturally evolved complex systems rather than an artifact of spectral analysis. Second, in the atmospheric domain, the framework naturally discriminates between structural failure modes (atmospheric blocking, where obsessive coherence is detected) and dynamic failure modes (storms, where it is not), demonstrating that the method detects a specific mechanism — crystallization preceding rupture — rather than generic anomaly.

These findings suggest that obsessive coherence may be a fundamental organizational principle of complex systems at criticality, analogous to how critical opalescence in physics signals a phase transition through excessive spatial correlation. If this interpretation is correct, the Kappa framework provides a domain-agnostic instrument for detecting the approach to structural failure in any system that can be represented as a network of coupled components.

Data and Code Availability

All code, data, and results are available at:

- **Kappa-SIG repository:** github.com/aprimora-ai/kappa-sig
- **Kappa-FIN (Sentinel):** github.com/aprimora-ai/heimdall
- **Kappa-Radiante:** github.com/aprimora-ai/kappa-radiante (DOI: 10.5281/zenodo.18940478)
- **Kappa Method:** DOI: 10.5281/zenodo.19339548 (v2.5)
- **Kappa-FIN:** DOI: 10.5281/zenodo.18917558
- **Kappa-LLM:** DOI: 10.5281/zenodo.18883790
- **Open-Meteo API:** open-meteo.com (atmospheric data, free access)
- **OULAD:** analyse.kmi.open.ac.uk/open_dataset (educational data, CC BY 4.0)
- **Political Blogosphere:** SNAP Stanford / Adamic & Glance (2005)
- **HaluEval:** [pminervini/HaluEval](https://pminervini.github.io/HaluEval) (HuggingFace Datasets)

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